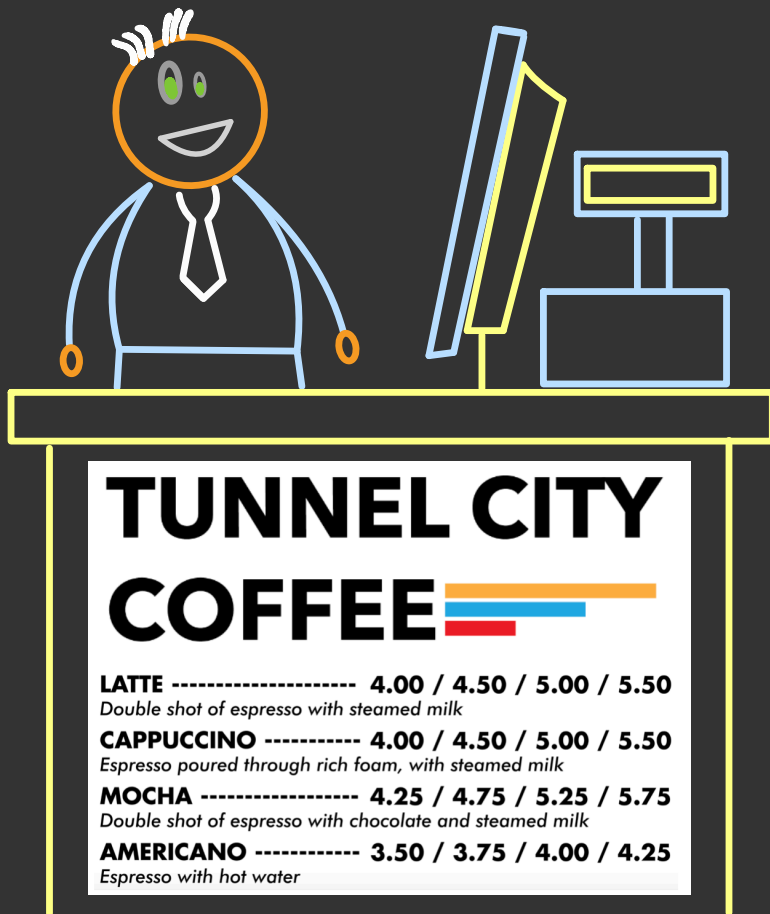
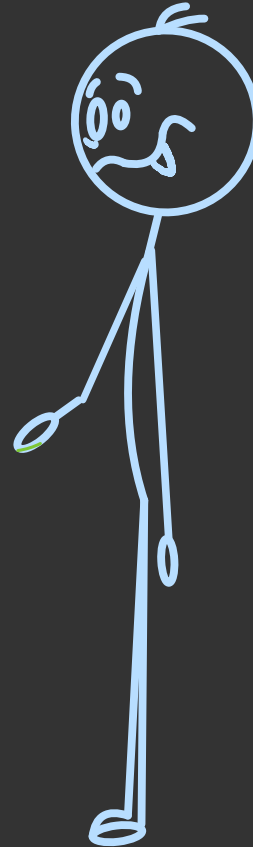
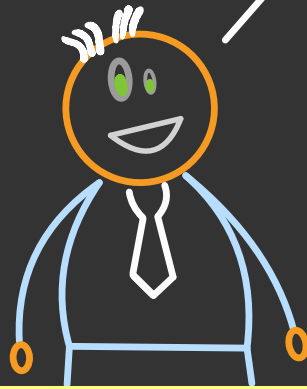


function graphs

CSCI
270



what size?



TUNNEL CITY COFFEE

LATTE ----- 4.00 / 4.50 / 5.00 / 5.50

Double shot of espresso with steamed milk

CAPPUCCINO ----- 4.00 / 4.50 / 5.00 / 5.50

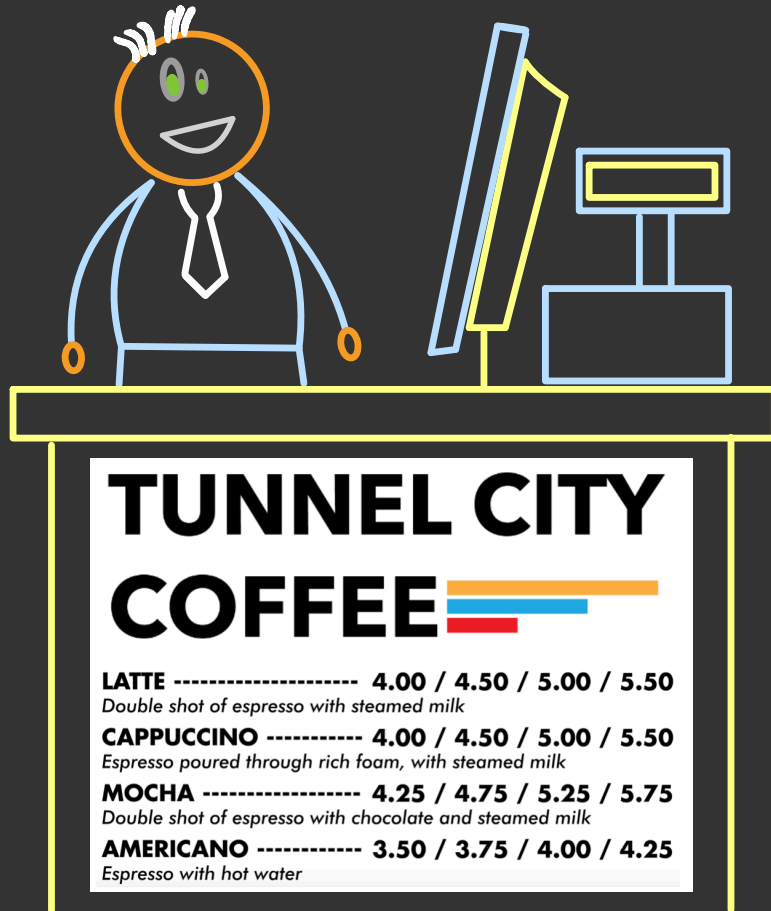
Espresso poured through rich foam, with steamed milk

MOCHA ----- 4.25 / 4.75 / 5.25 / 5.75

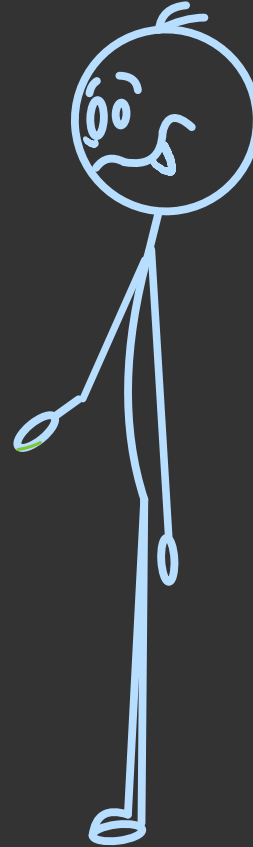
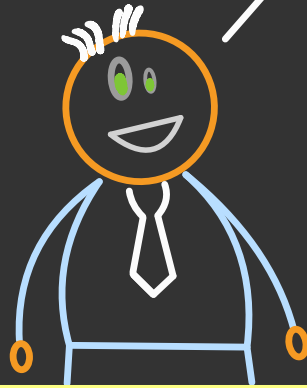
Double shot of espresso with chocolate and steamed milk

AMERICANO ----- 3.50 / 3.75 / 4.00 / 4.25

Espresso with hot water



single or double shot?



TUNNEL CITY COFFEE

LATTE ----- 4.00 / 4.50 / 5.00 / 5.50

Double shot of espresso with steamed milk

CAPPUCCINO ----- 4.00 / 4.50 / 5.00 / 5.50

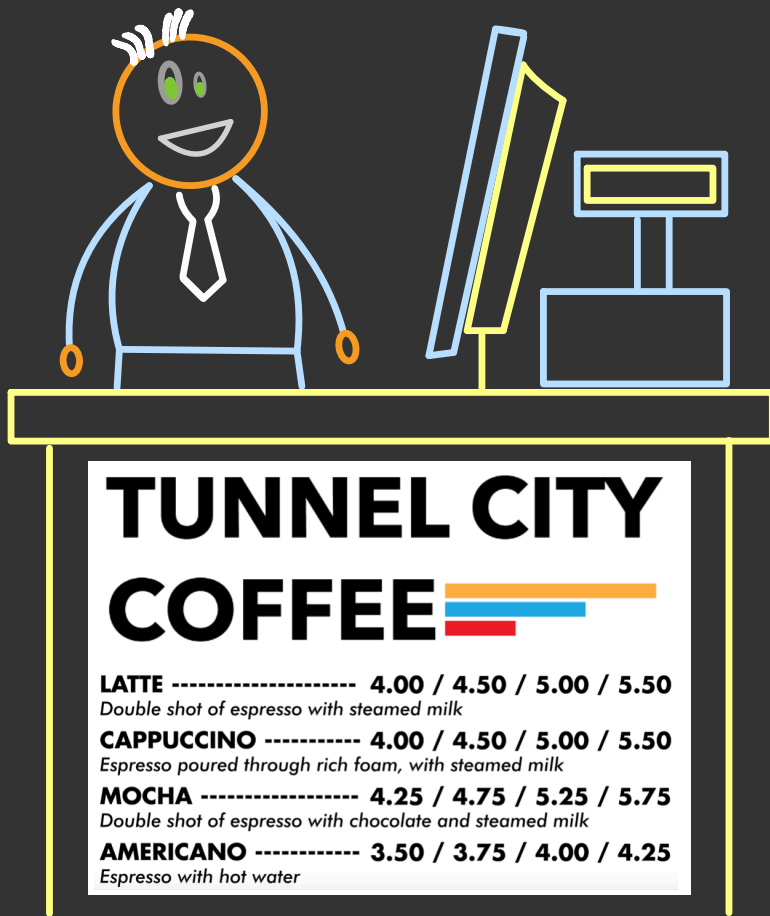
Espresso poured through rich foam, with steamed milk

MOCHA ----- 4.25 / 4.75 / 5.25 / 5.75

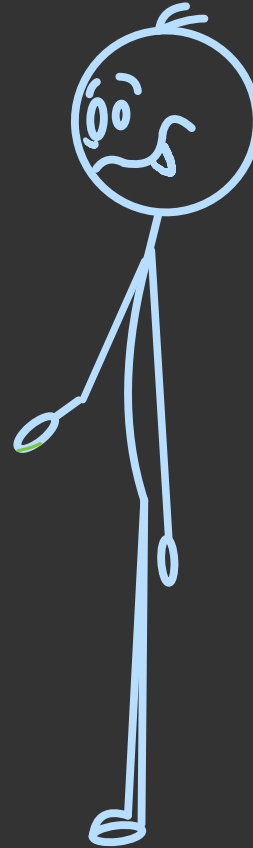
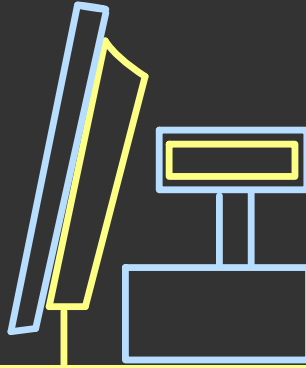
Double shot of espresso with chocolate and steamed milk

AMERICANO ----- 3.50 / 3.75 / 4.00 / 4.25

Espresso with hot water



what kind of milk?



TUNNEL CITY COFFEE

LATTE ----- 4.00 / 4.50 / 5.00 / 5.50

Double shot of espresso with steamed milk

CAPPUCCINO ----- 4.00 / 4.50 / 5.00 / 5.50

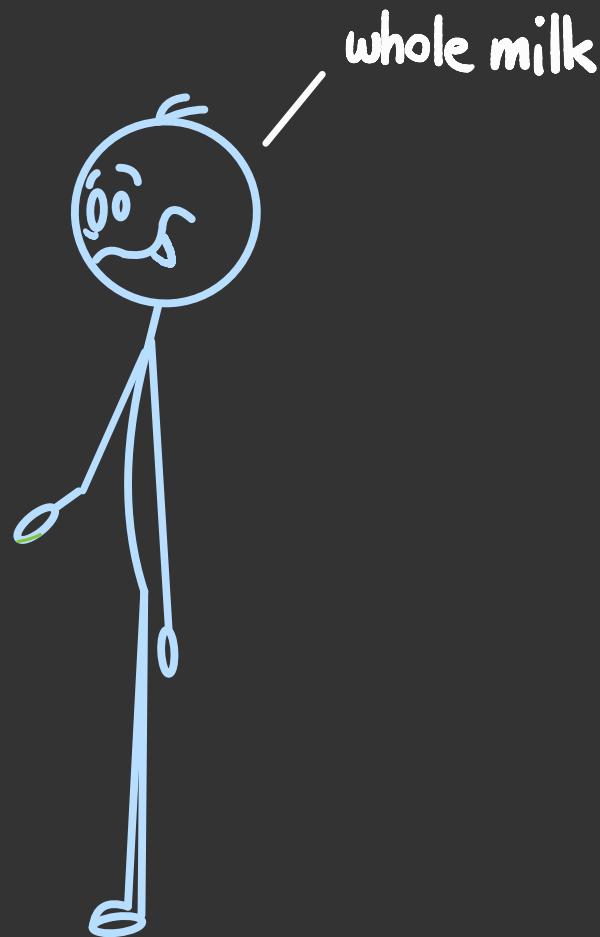
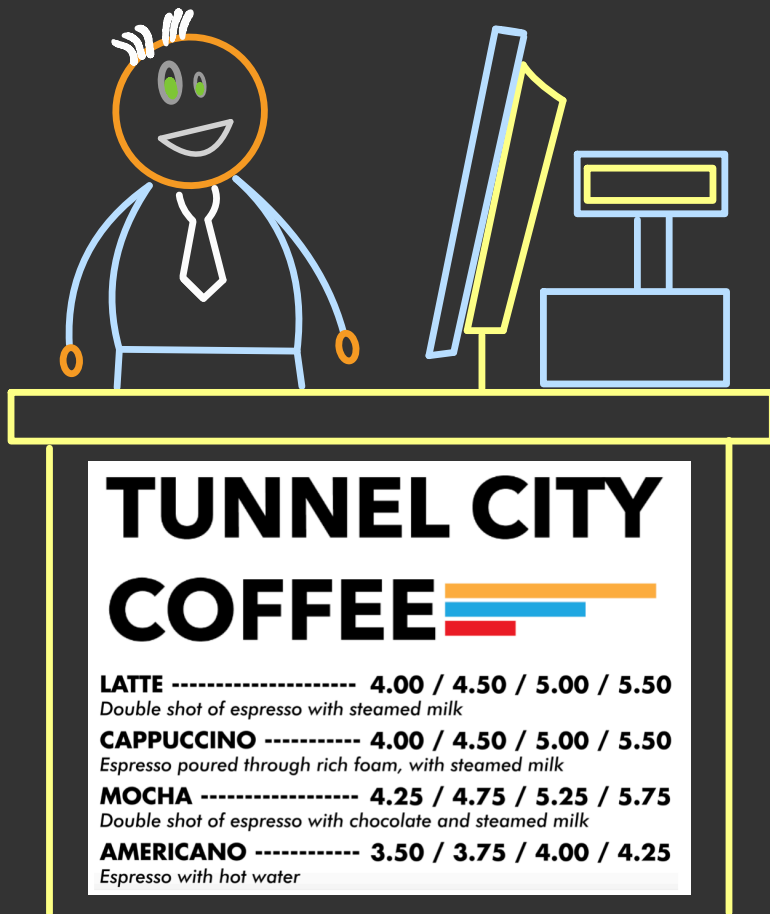
Espresso poured through rich foam, with steamed milk

MOCHA ----- 4.25 / 4.75 / 5.25 / 5.75

Double shot of espresso with chocolate and steamed milk

AMERICANO ----- 3.50 / 3.75 / 4.00 / 4.25

Espresso with hot water





oat milk?

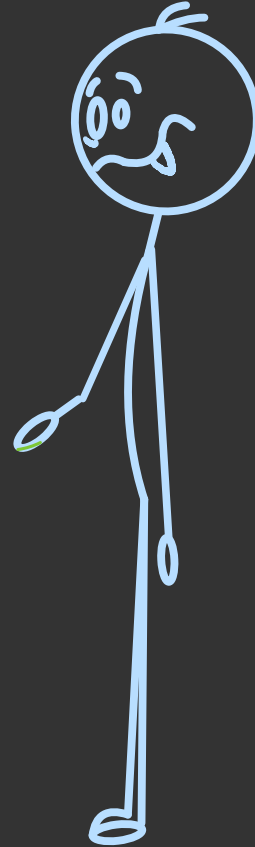
TUNNEL CITY COFFEE

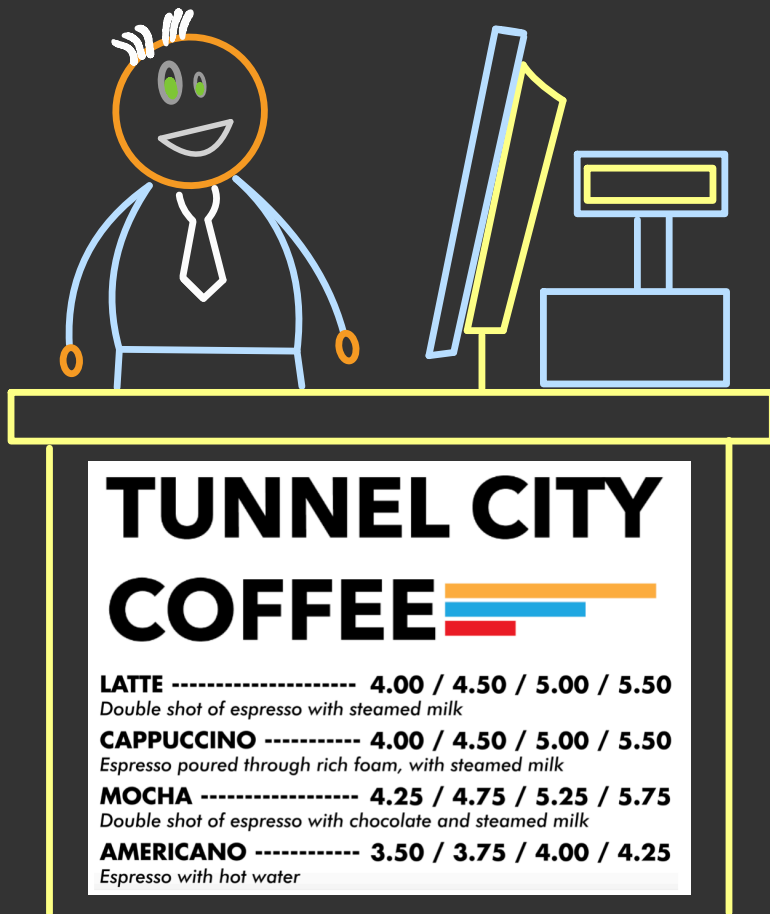
LATTE ----- 4.00 / 4.50 / 5.00 / 5.50
Double shot of espresso with steamed milk

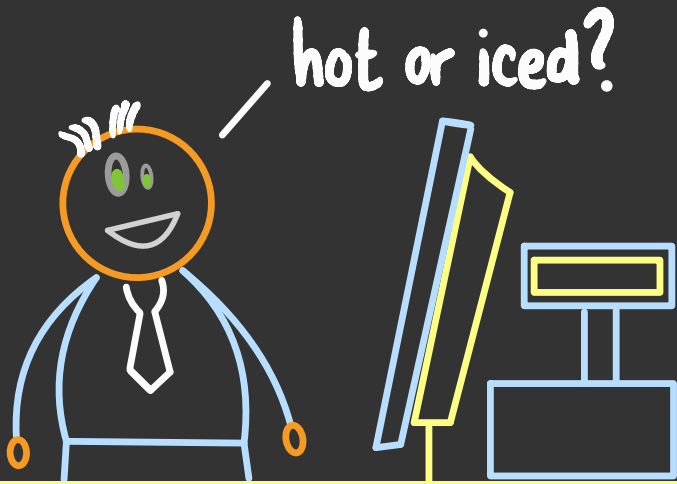
CAPPUCCINO ----- 4.00 / 4.50 / 5.00 / 5.50
Espresso poured through rich foam, with steamed milk

MOCHA ----- 4.25 / 4.75 / 5.25 / 5.75
Double shot of espresso with chocolate and steamed milk

AMERICANO ----- 3.50 / 3.75 / 4.00 / 4.25
Espresso with hot water



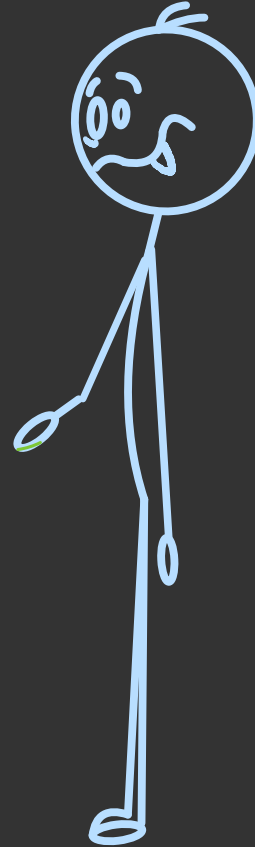


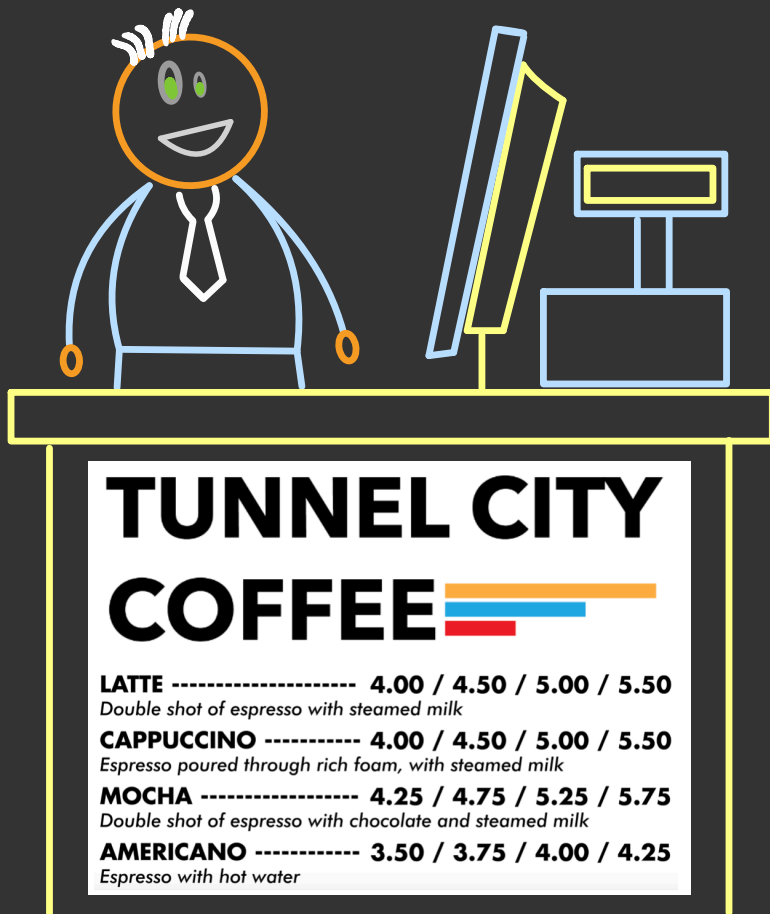


hot or iced?

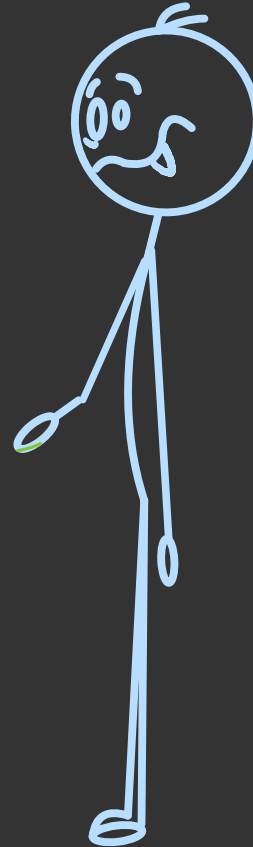
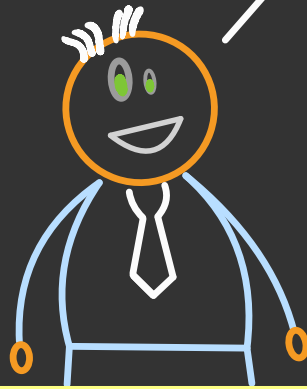
TUNNEL CITY COFFEE

LATTE -----	4.00 / 4.50 / 5.00 / 5.50
<i>Double shot of espresso with steamed milk</i>	
CAPPUCCINO -----	4.00 / 4.50 / 5.00 / 5.50
<i>Espresso poured through rich foam, with steamed milk</i>	
MOCHA -----	4.25 / 4.75 / 5.25 / 5.75
<i>Double shot of espresso with chocolate and steamed milk</i>	
AMERICANO -----	3.50 / 3.75 / 4.00 / 4.25
<i>Espresso with hot water</i>	





what size?



TUNNEL CITY COFFEE

LATTE ----- 4.00 / 4.50 / 5.00 / 5.50

Double shot of espresso with steamed milk

CAPPUCCINO ----- 4.00 / 4.50 / 5.00 / 5.50

Espresso poured through rich foam, with steamed milk

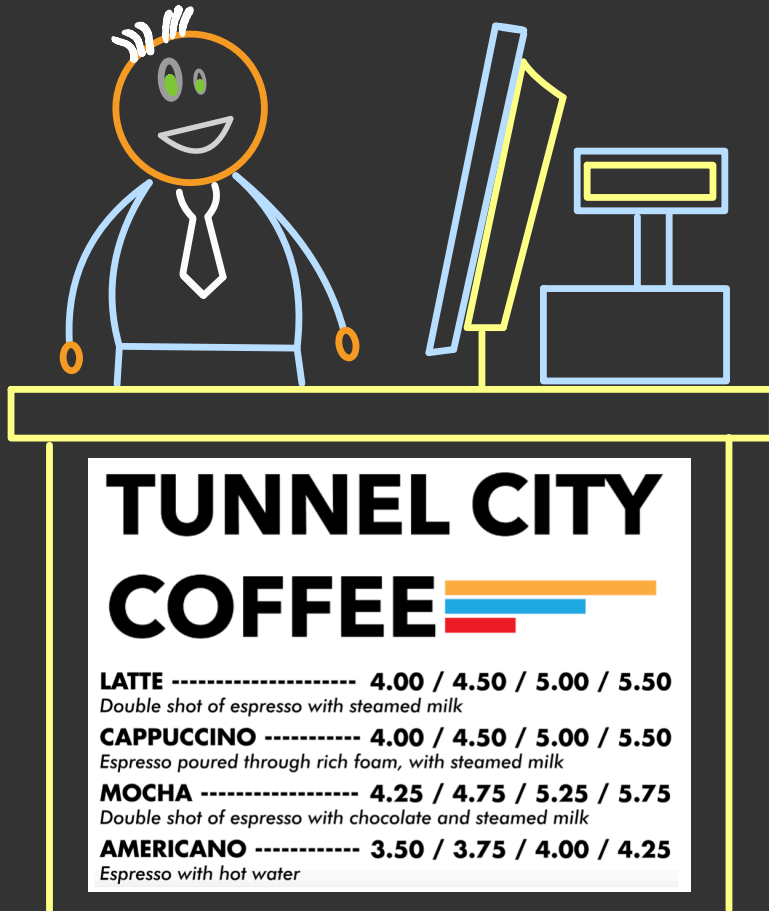
MOCHA ----- 4.25 / 4.75 / 5.25 / 5.75

Double shot of espresso with chocolate and steamed milk

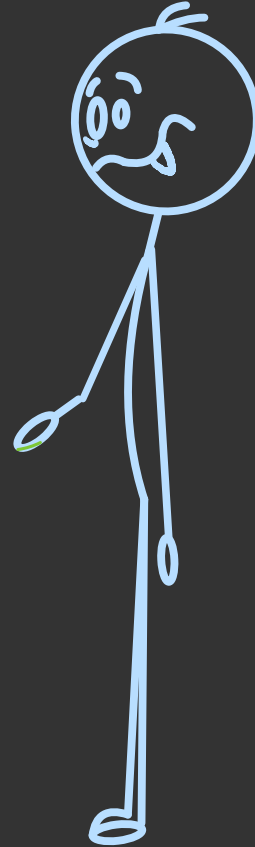
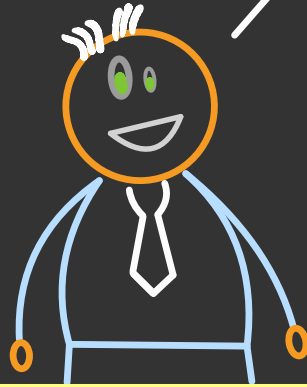
AMERICANO ----- 3.50 / 3.75 / 4.00 / 4.25

Espresso with hot water

a little while
later



that will be \$4.78



TUNNEL CITY COFFEE

LATTE ----- 4.00 / 4.50 / 5.00 / 5.50

Double shot of espresso with steamed milk

CAPPUCCINO ----- 4.00 / 4.50 / 5.00 / 5.50

Espresso poured through rich foam, with steamed milk

MOCHA ----- 4.25 / 4.75 / 5.25 / 5.75

Double shot of espresso with chocolate and steamed milk

AMERICANO ----- 3.50 / 3.75 / 4.00 / 4.25

Espresso with hot water

Sales tax

The Massachusetts sales tax is **6.25%** of the sales price or rental charge of tangible personal property (including gas, electricity, and steam) and telecommunications services¹ sold or rented in Massachusetts. The buyer pays the sales tax, as an addition to the purchase price, to the vendor at the time of purchase. The vendor then sends the tax to Massachusetts.

6.25%

of the sales price or rental charge

TUNNEL CITY COFFEE

LATTE ----- **4.00 / 4.50 / 5.00 / 5.50**

Double shot of espresso with steamed milk

CAPPUCCINO ----- **4.00 / 4.50 / 5.00 / 5.50**

Espresso poured through rich foam, with steamed milk

MOCHA ----- **4.25 / 4.75 / 5.25 / 5.75**

Double shot of espresso with chocolate and steamed milk

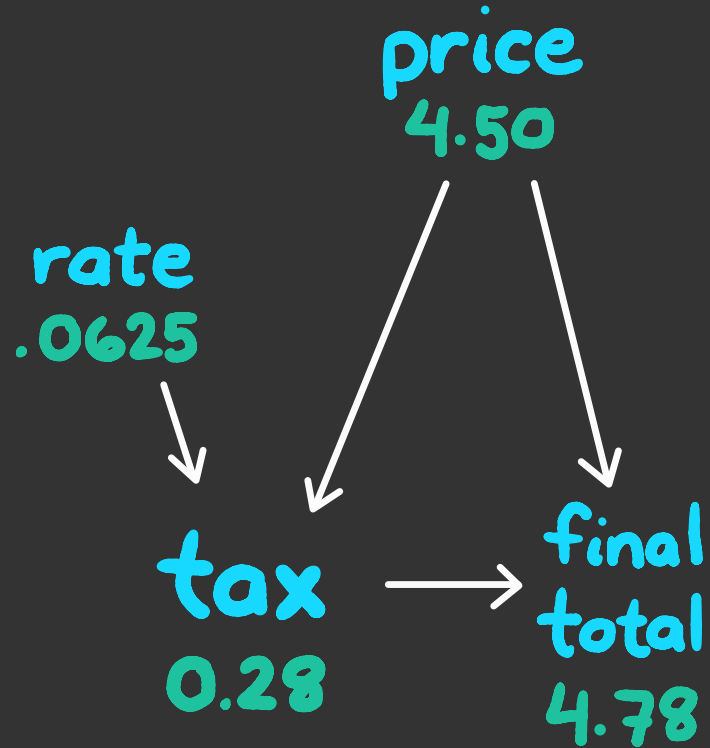
AMERICANO ----- **3.50 / 3.75 / 4.00 / 4.25**

Espresso with hot water

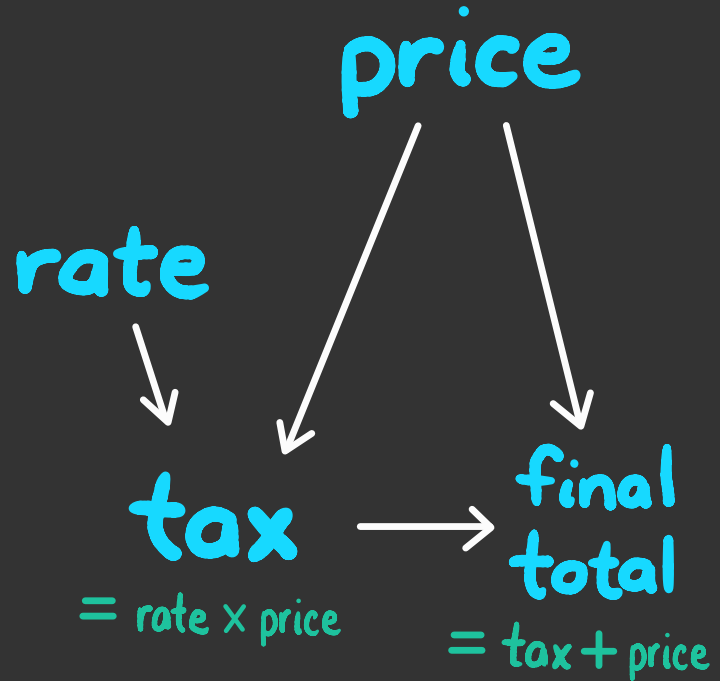
$$.0625 \times 4.50$$

$$\approx 0.28$$

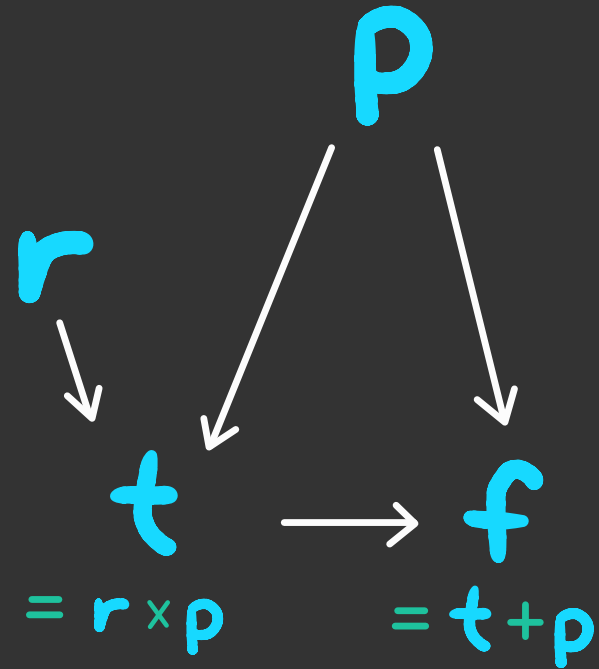
we can represent
this computation
as a graph



each vertex is
a function of
its parents

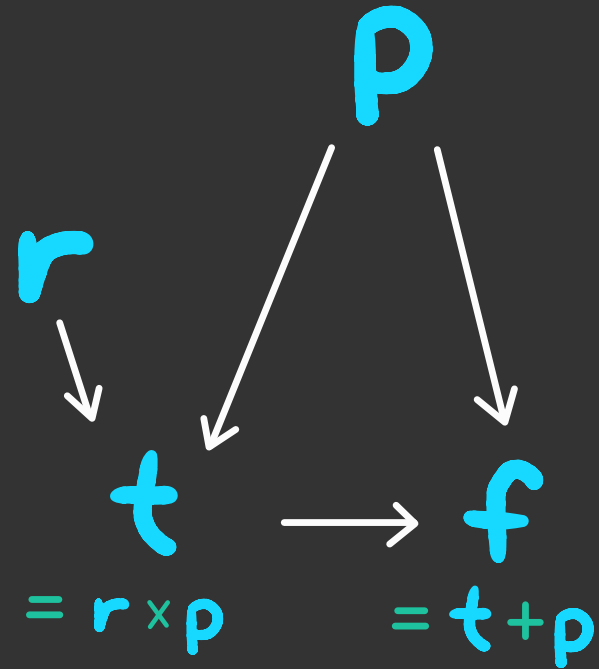


each vertex is
a function of
its parents



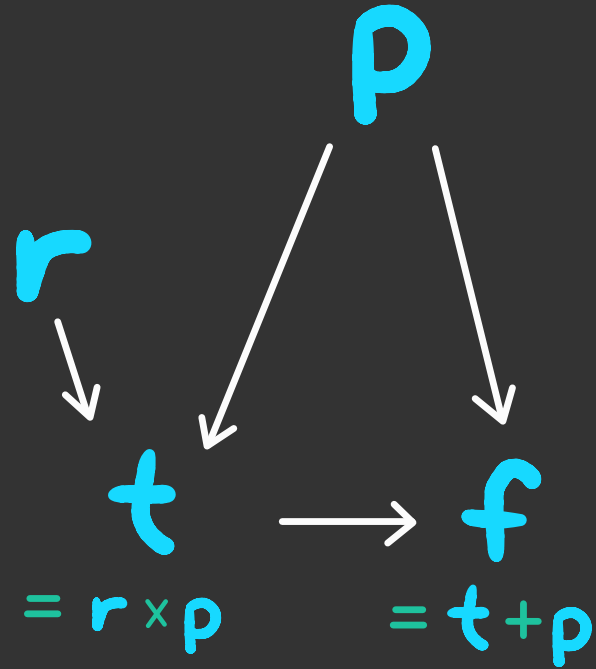
how does the
final total
change

as we increase
the tax **rate**?

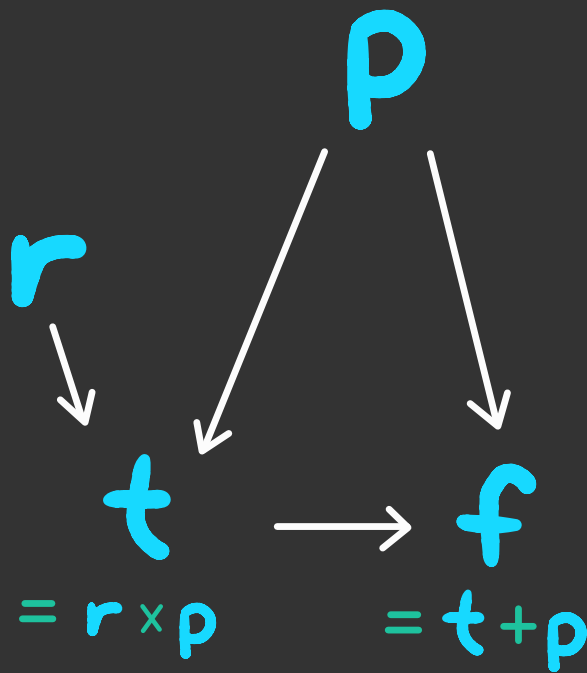


how does the
final total
change as we increase
the tax **rate**?

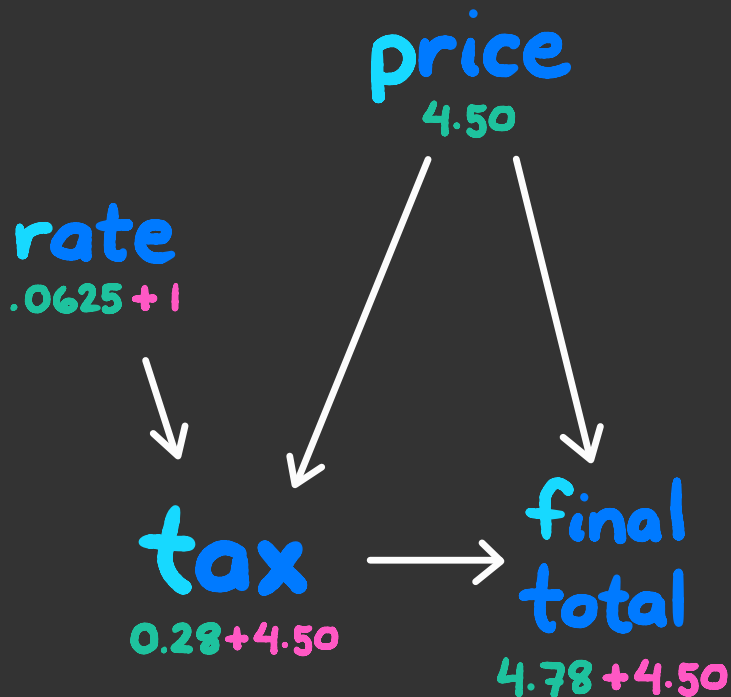
$$\frac{\partial f}{\partial r} = ?$$



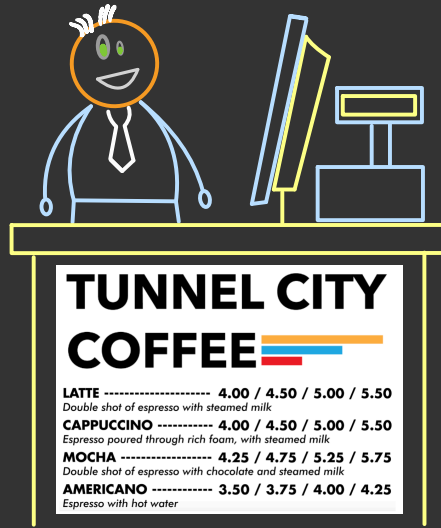
$$\begin{aligned}
 \frac{\partial f}{\partial r} &= \frac{\partial}{\partial r} (t+p) \\
 &= \frac{\partial}{\partial r} (rp+p) \\
 &= \frac{\partial}{\partial r} (rp) + \frac{\partial}{\partial r} (p) \\
 &= p + 0 \\
 &= p
 \end{aligned}$$



$$\frac{\partial f}{\partial r} = P$$

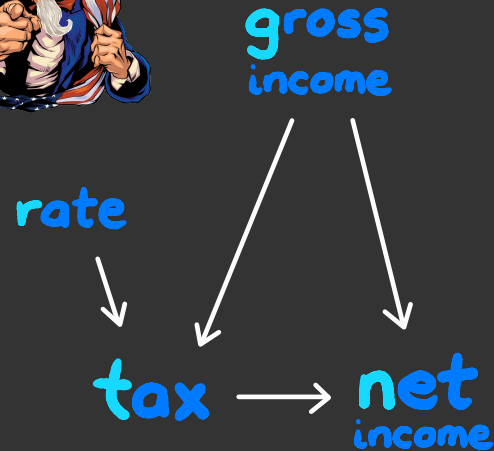
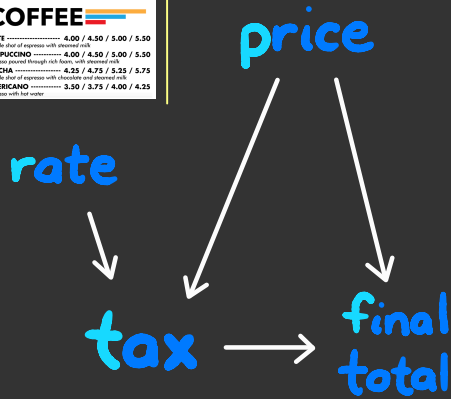
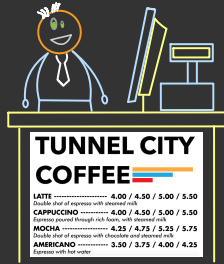


so that's
sales tax

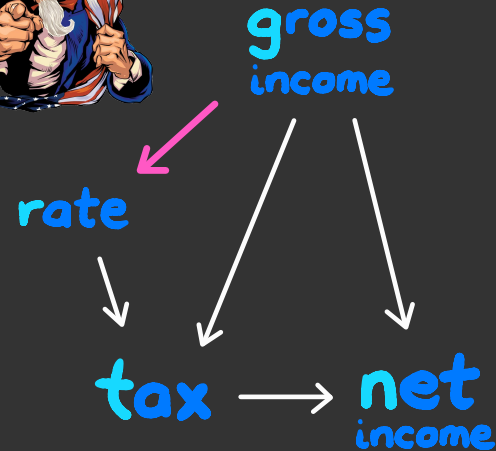
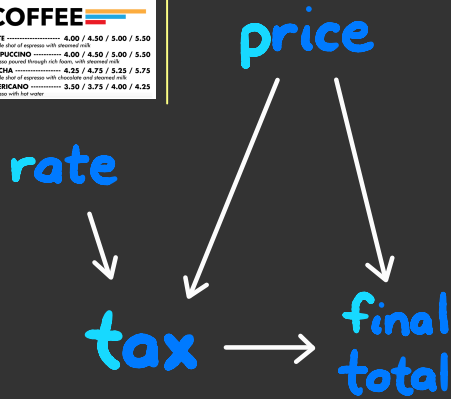
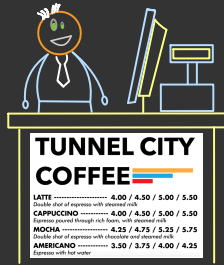


what about
income tax?

is it the same graph?

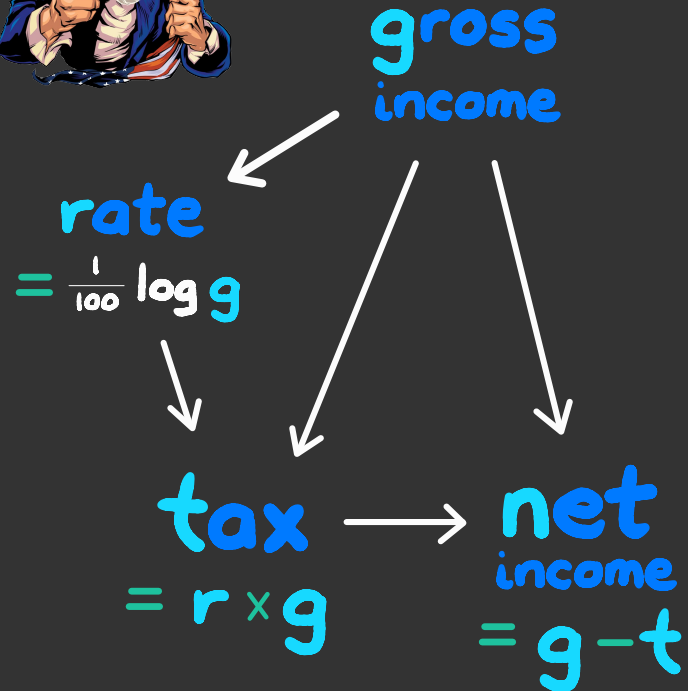


not if the tax rate depends on your income



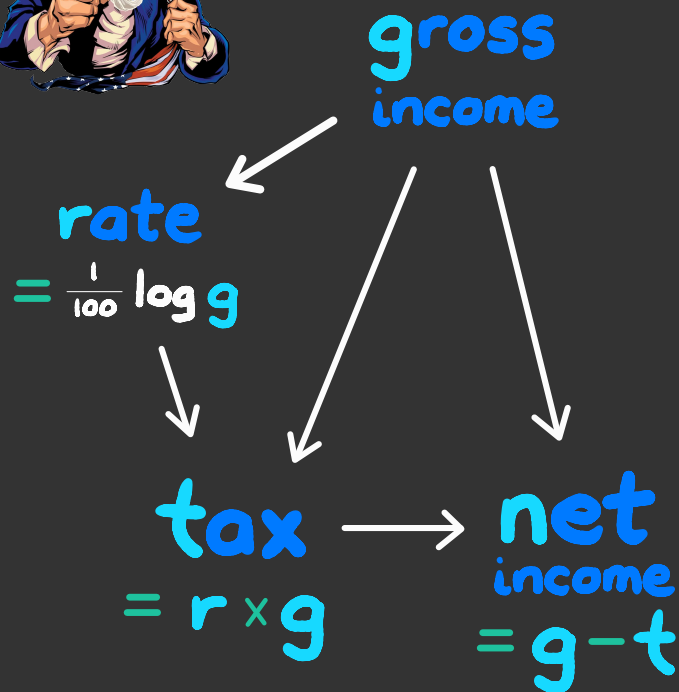
how does
net income
change as we increase
gross income?

$$\frac{\partial n}{\partial g} = ?$$



$$\frac{\partial n}{\partial g} = \frac{\partial}{\partial g} (g - t)$$

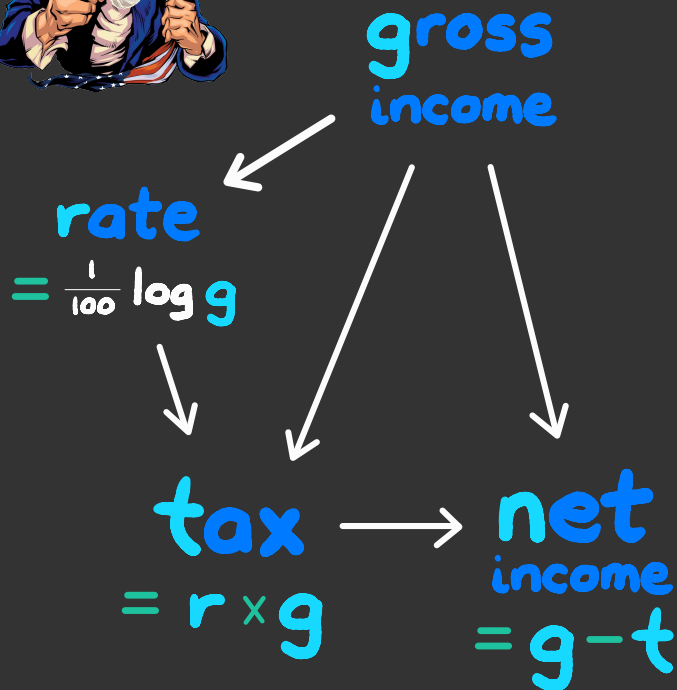
$$= 1 \text{ ?}$$



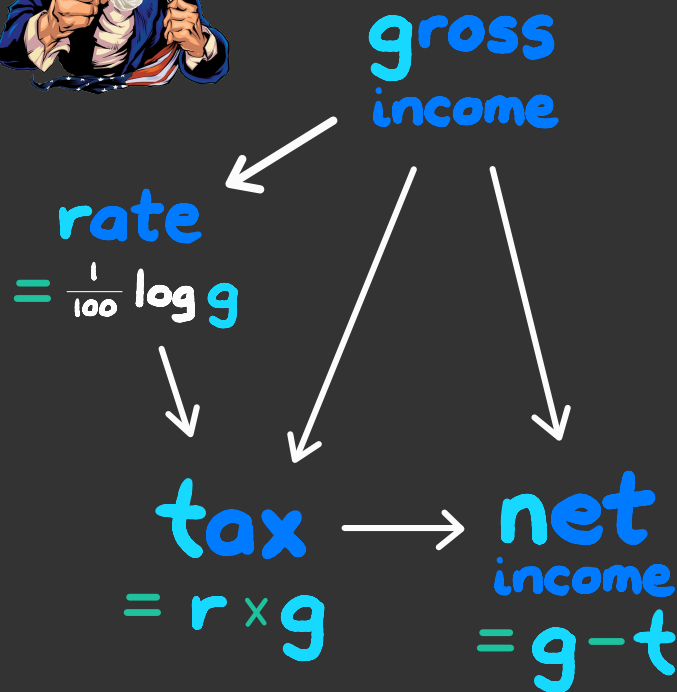
$$\frac{\partial n}{\partial g} = \frac{\partial}{\partial g} (g - t)$$

$$= 1 \quad ?$$

no, since t
depends on g



$$\begin{aligned}
 \frac{\partial n}{\partial g} &= \frac{\partial}{\partial g} (g - t) \\
 &= \frac{\partial}{\partial g} (g - rg) \\
 &= 1 - r \text{ ?}
 \end{aligned}$$

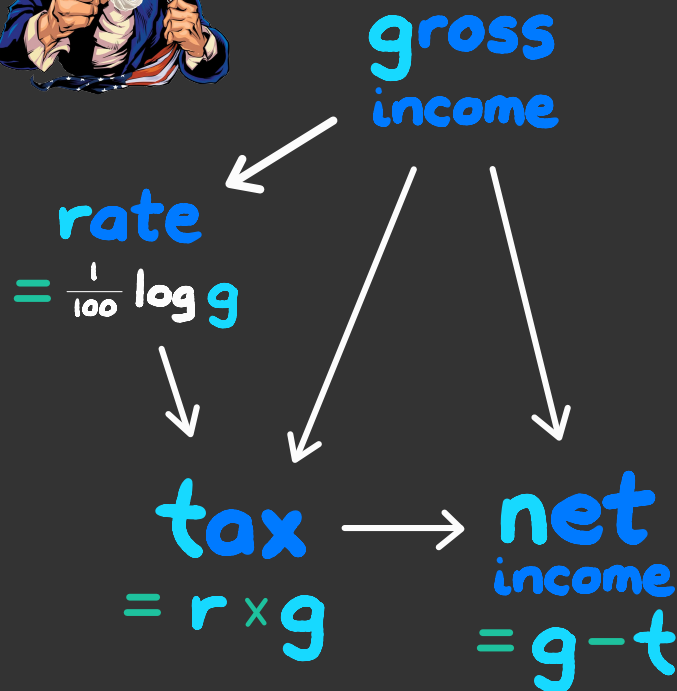


$$\frac{\partial n}{\partial g} = \frac{\partial}{\partial g} (g - t)$$

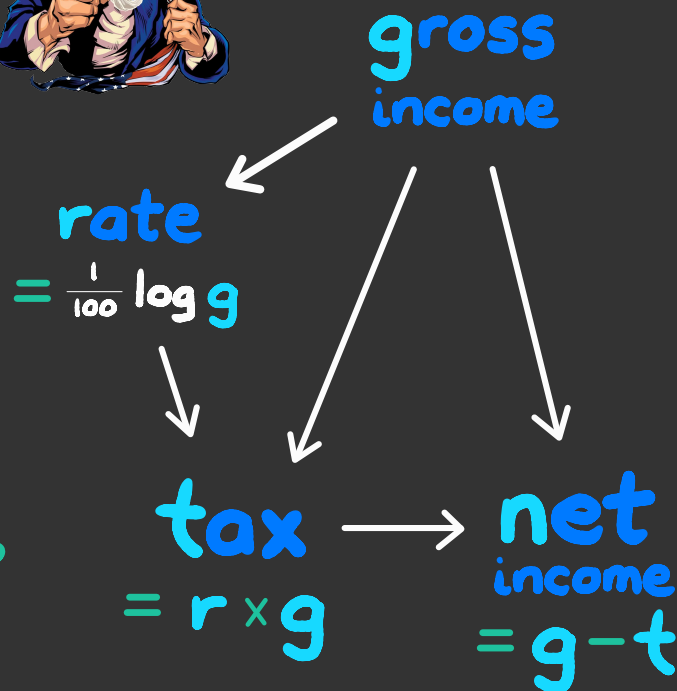
$$= \frac{\partial}{\partial g} (g - rg)$$

$$= 1 - r$$

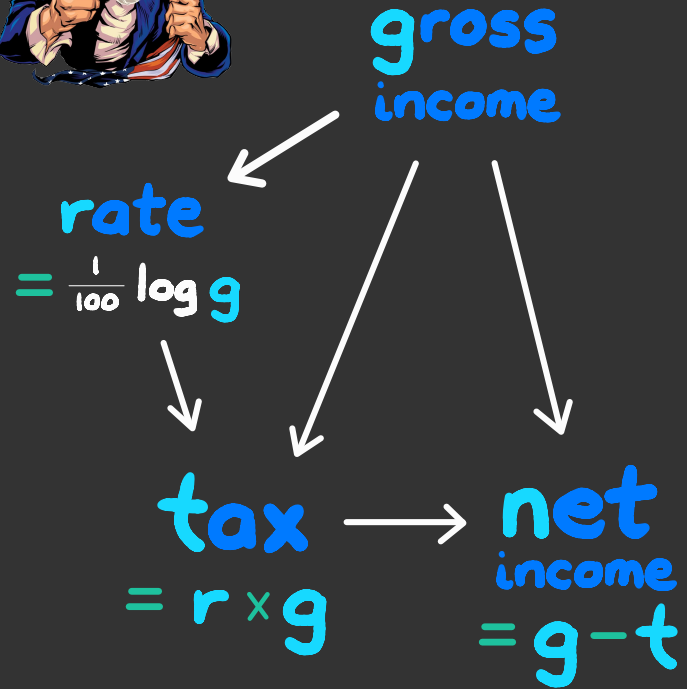
• no, since r depends on g



$$\begin{aligned}
 \frac{\partial n}{\partial g} &= \frac{\partial}{\partial g} (g - t) \\
 &= \frac{\partial}{\partial g} (g - rg) \\
 &= \frac{\partial}{\partial g} \left(g - g \left(\frac{1}{100} \log g \right) \right) \\
 &= 1 - \frac{\log g}{100} - \frac{1}{100} \quad \checkmark
 \end{aligned}$$

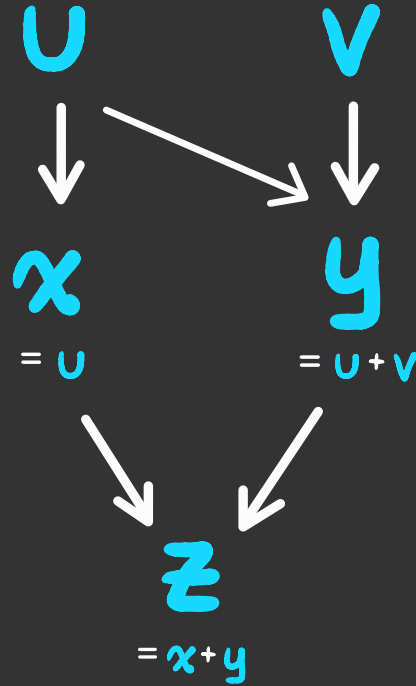


as the graph gets
more complicated,
backsubstitution
can get computationally
expensive

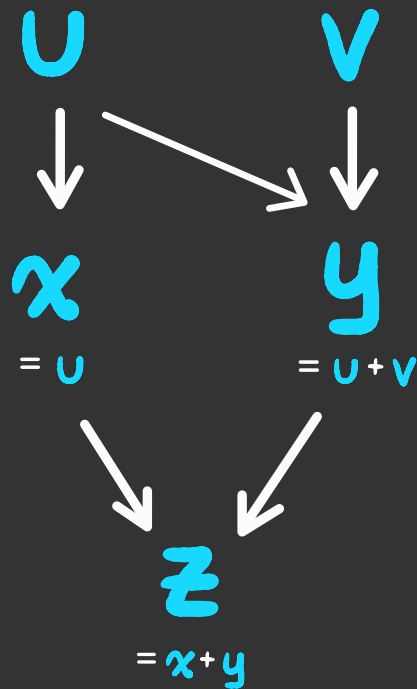


is there a **better** way?

consider this
function
graph



$$\frac{\partial z}{\partial x} = ?$$

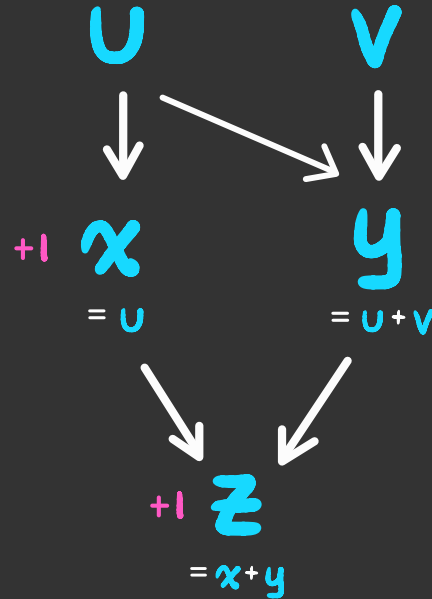


variables depend only on their ancestors in the function graph

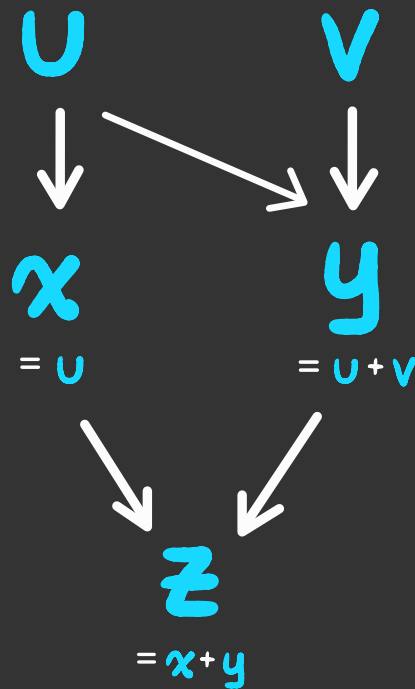
$$\frac{\partial z}{\partial x} = \frac{\partial}{\partial x} (x + y)$$

$= 1$

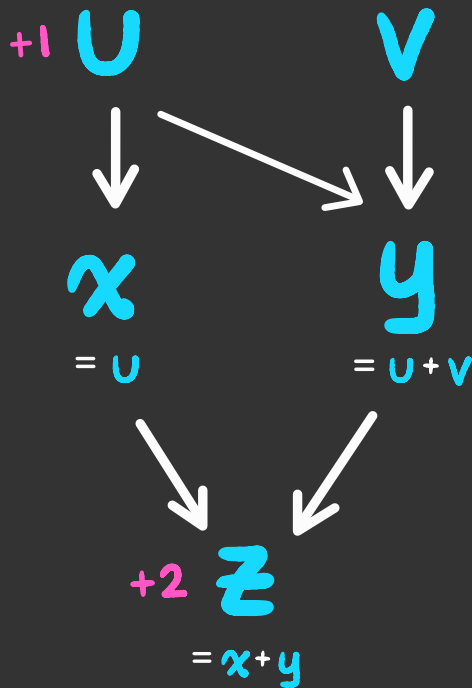
doesn't depend on x



$$\frac{\partial z}{\partial u} = ?$$



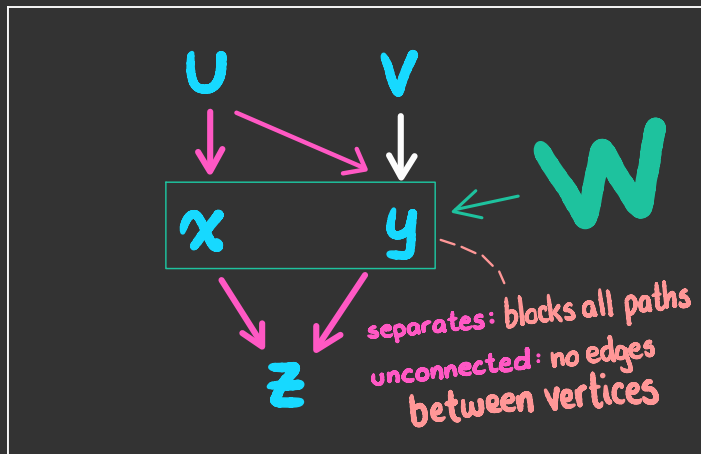
$$\begin{aligned}
 \frac{\partial z}{\partial u} &= \frac{\partial}{\partial u} (x + y) \\
 &= \frac{\partial}{\partial u} (u + u + v) \\
 &= 2
 \end{aligned}$$



here's the
better way

the chain rule of partial derivatives

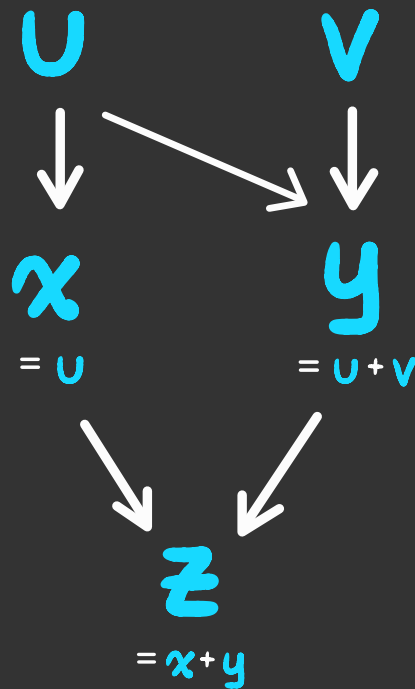
suppose a pairwise
unconnected vertex
set W separates
vertex u
and vertex z
in a function graph



then

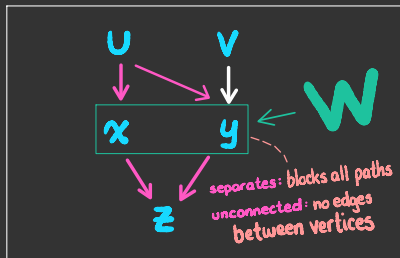
$$\frac{\partial z}{\partial u} = \sum_{w \in W} \frac{\partial z}{\partial w} \frac{\partial w}{\partial u}$$

$$\frac{\partial z}{\partial u} = ?$$

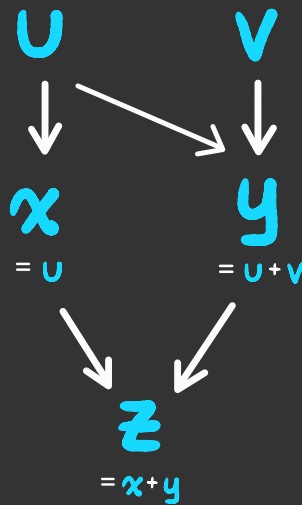


the chain rule of partial derivatives

suppose a pairwise
unconnected vertex
set W separates
vertex u
and vertex z
in a function graph



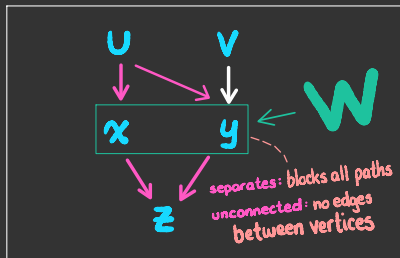
then
$$\frac{\partial z}{\partial u} = \sum_{w \in W} \frac{\partial z}{\partial w} \frac{\partial w}{\partial u}$$



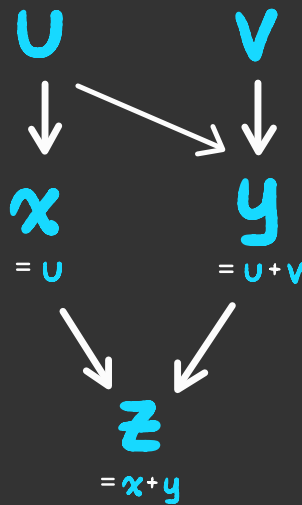
$$\frac{\partial z}{\partial u} = ?$$

the chain rule of partial derivatives

suppose a pairwise
unconnected vertex
set W separates
vertex u
and vertex z
in a function graph



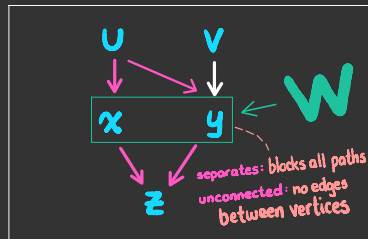
then
$$\frac{\partial z}{\partial u} = \sum_{w \in W} \frac{\partial z}{\partial w} \frac{\partial w}{\partial u}$$



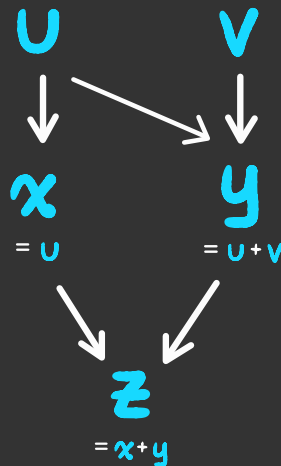
$$\frac{\partial z}{\partial u} = ?$$

the chain rule of partial derivatives

suppose a pairwise
unconnected vertex
set W separates
vertex u
and vertex z
in a function graph



$$\text{then } \frac{\partial z}{\partial u} = \sum_{w \in W} \frac{\partial z}{\partial w} \frac{\partial w}{\partial u}$$



$$\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u}$$

$$= \frac{\partial}{\partial x}(x+y) \frac{\partial}{\partial u}(u) + \frac{\partial}{\partial y}(x+y) \frac{\partial}{\partial u}(u+v)$$

$$= 1 \cdot 1 + 1 \cdot 1$$

$$= 2$$